

therefore.

$$\tilde{X}Y:YO = YX:XO$$

$$\Leftrightarrow |\tilde{x}-y|:|y| = |x-y|:|x|$$

$$\Leftrightarrow |x| \cdot |\tilde{x}-y| = |y| \cdot |x-y|$$

$$\therefore |x| \cdot |\tilde{x}-y| = |x-y| \quad (\because y \in \partial B_1(o))$$

$$\text{Thus } \Phi(y-x) = \Phi(|x|(\tilde{x}-y)) \quad (\because \forall y \in \partial \Omega = \partial B_1(o)) \\ = \Phi(|x|(y-\tilde{x})).$$

Since

$$\Delta_y \Phi(|x|(y-\tilde{x})) = 0 \quad \forall y \in B_1(o)$$

$$\phi^x(y) = \Phi(|x|(y-\tilde{x})) \text{ is a sol of (11.6)}$$

$$\Phi(|x|(y-\tilde{x})) = \begin{cases} -\frac{1}{2\pi} \log(|x| \cdot |y-\tilde{x}|) & ; n=2 \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{(|x| \cdot |y-\tilde{x}|)^{n-2}} & ; n \geq 3 \end{cases}$$

Thm 11.3

The Green func. of (11.1) with $\Omega = B_1(o)$

can be represented as

$$G(x,y) = \begin{cases} -\frac{1}{2\pi} \left(\log|x-y| + \log \left(|x| \cdot \left| y - \frac{x}{|x|^2} \right| \right) \right) & ; n=2 \\ \frac{1}{n(n-2)\alpha(n)} \left(\frac{1}{|x-y|^{n-2}} + \frac{1}{|x| \cdot \left| y - \frac{x}{|x|^2} \right|^{n-2}} \right) & n \geq 3 \end{cases}$$

Let $g: \partial B_1(o) \rightarrow \mathbb{R}$ be a given continuous func. Consider

$$\begin{cases} -\Delta u(x) = 0, & x \in B_1(o) \\ u(x) = g(x); & x \in \partial B_1(o) \end{cases} \quad (11.8)$$

By Thm 11.1 or 11.2

$$u(y) = - \int_{\partial B_1(o)} g(y) \nabla_y G(x,y) \cdot \nu(y) d\tau(y).$$

By direct computation of $\nabla_y G(x,y)$, we have

$$u(y) = \frac{1-|x|^2}{n\alpha(n)} \int_{\partial B_1(o)} \frac{g(y)}{|x-y|^n} d\tau(y)$$

(Poisson Integral)

(ref. Gilberg - Trudinger (1982) or reprint ver.).

Chapter 12: Fundamental sol for heat eq.

$$(12.1) \quad \frac{\partial u}{\partial t}(x,t) - \Delta u(x,t) = f(x,t), \quad x \in \Omega, t > 0$$

Fundamental sol.

Let $\Omega = \mathbb{R}^n$ and $f \equiv 0$.

Prop 12.1: Let $\Omega = \mathbb{R}^n$ and $f \equiv 0$. Let u be a sol of (12.1)

For $\lambda > 0$, let

$$(12.2) \quad \tilde{u}(\tilde{x}, \tilde{t}) = u(x, t), \quad x = \lambda \tilde{x}, \tilde{t} = \lambda^2 t$$

$\Rightarrow \tilde{u}$ is a sol of (12.1)

$$\text{i.e. } \frac{\partial \tilde{u}}{\partial \tilde{t}}(\tilde{x}, \tilde{t}) - \Delta_{\tilde{x}} \tilde{u}(\tilde{x}, \tilde{t}) = 0, \quad \tilde{x} \in \mathbb{R}^n, \tilde{t} > 0$$

$$\text{pf. } \frac{\partial}{\partial \tilde{t}} \tilde{u}(\tilde{x}, \tilde{t}) \stackrel{(12.2)}{=} \frac{\partial}{\partial \tilde{t}} u(x, t)$$

$$= \frac{\partial}{\partial \tilde{t}} u \left(\frac{\tilde{x}}{\lambda}, \frac{\tilde{t}}{\lambda^2} \right)^{\lambda^2 t} \quad (x = \lambda \tilde{x}, \tilde{t} = \lambda^2 t)$$

$$= \frac{\partial u}{\partial t} \left(\frac{\tilde{x}}{\lambda}, \frac{\tilde{t}}{\lambda^2} \right) \left(\frac{\tilde{t}}{\lambda^2} \right)^{\lambda^2 t}$$

$$= \frac{\partial u}{\partial t} \left(\boxed{\tilde{x}}, \lambda \tilde{t} \right) \lambda^2$$

$$\therefore \frac{\partial \tilde{u}}{\partial \tilde{t}}(\tilde{x}, \tilde{t}) = \lambda^2 \frac{\partial u}{\partial t}(x, t)$$

Similarly $\frac{\partial \tilde{u}}{\partial \tilde{x}_j}(\tilde{x}, \tilde{t}) = \lambda \frac{\partial u}{\partial x_j}(x, t)$

$$\frac{\partial^2 \tilde{u}}{\partial \tilde{x}_j \partial \tilde{x}_i}(\tilde{x}, \tilde{t}) = \lambda^2 \frac{\partial^2 u}{\partial x_i \partial x_j}(x, t)$$

Since u is a sol of (12.1)

$$\frac{\partial \tilde{u}}{\partial \tilde{t}}(\tilde{x}, \tilde{t}) = \lambda^2 \frac{\partial u}{\partial t}(x, t)$$

$$= \lambda^2 \Delta u(x, t)$$

$$= \Delta_{\tilde{x}} \tilde{u}(\tilde{x}, \tilde{t}) \quad \square$$

$$x = \lambda \tilde{x}, \quad t = \lambda^2 \tilde{t}$$

$$\Leftrightarrow \lambda = x/\tilde{x}, \quad t = \lambda^2 \tilde{t}$$

$$\Rightarrow t = \left(\frac{x}{\tilde{x}}\right)^2 \tilde{t} \Leftrightarrow \frac{\tilde{x}^2}{\tilde{t}} = \frac{x^2}{t}$$

Put $y = x/\sqrt{t}$ and find a sol of (12.1) of the form.

$$(12.3) \quad u(x, t) = t^\alpha v(y), \quad y = x/\sqrt{t}$$

Compare $\partial u / \partial t(x, t)$:

$$\frac{\partial u}{\partial t}(x, t) = \frac{\partial}{\partial t} u(x, t) = \frac{\partial}{\partial t} (t^\alpha v(y))$$

$$= \frac{\partial}{\partial t} \left(t^\alpha v\left(\frac{x}{\sqrt{t}}\right) \right)$$

$$= \alpha t^{\alpha-1} v\left(\frac{x}{\sqrt{t}}\right) + t^\alpha \nabla_y \left(\frac{x}{\sqrt{t}}\right) \times \left(\frac{x}{\sqrt{t}}\right)_t$$

$$= \alpha t^{\alpha-1} v(y) - \frac{1}{2} t^{\alpha-3/2} \nabla_y v(y) \cdot y$$

$$= \alpha t^{\alpha-1} v(y) - \frac{1}{2} t^{\alpha-1/2} \nabla_y v(y) \cdot y$$

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Compare $\frac{\partial u}{\partial x_i}(x, t)$:

$$\begin{aligned}
 \frac{\partial u}{\partial x_i}(x, t) &= \frac{\partial}{\partial x_i} u(x, t) = \frac{\partial}{\partial x_i} (t^\alpha v(y)) \\
 &= \frac{\partial}{\partial x_i} \left(t^\alpha v\left(\frac{x}{\sqrt{t}}\right) \right) \\
 &= t^\alpha \frac{\partial v}{\partial y_i} \left(\frac{x}{\sqrt{t}} \right) \cdot \frac{1}{\sqrt{t}} \quad (\text{chain rule}) \\
 &= t^{\alpha - \frac{1}{2}} \frac{\partial v}{\partial y_i}(y)
 \end{aligned}$$

Similarity Compare $\partial^2 u / \partial x_i \partial x_i \cdot (y)$:

$$\frac{\partial^2 u}{\partial x_i \partial x_i}(x, t) = t^{\alpha - 1} \frac{\partial^2 u}{\partial x_i \partial x_i}(y)$$

Plugging them into (12.1) we have

$$\alpha t^{\alpha - 1} v(y) - \frac{1}{2} t^{\alpha - 1} \nabla_y v(y) \cdot y = t^{\alpha - 1} \Delta_y v(y)$$

$$\Rightarrow \alpha v(y) - \frac{1}{2} \nabla_y v(y) \cdot y - \Delta_y v(y) = 0.$$

Let us assume v is radially symmetric:

$$v(y) = w(r), \quad r = |y|$$

Then

$$\nabla_y v(y) = \frac{y}{r} w_r(r).$$

$$\Delta_y v(y) = w_{rr}(r) + \frac{n-1}{r} w_r(r) \quad (\text{Exercise}).$$

hence

$$\alpha w(r) - \frac{1}{2} \left(\frac{y}{r} w_r(r) \right) \cdot y - w_{rr}(r) - \frac{n-1}{r} w_r(r) = 0$$

$$\frac{w_r(r)}{r} y \cdot y = \frac{w_r(r)}{r} |y|^2 = r w_r(r)$$

$$\Rightarrow \alpha w(r) - \frac{1}{2} r w_r(r) - \left(w_{rr}(r) + \frac{n-1}{r} w_r(r) \right) = 0.$$

$$\begin{aligned} \text{multiply } r^{n-1} \Rightarrow & \alpha r^{n-1} w(r) - \frac{1}{2} r w_r(r) - \left(r^{n-1} w_{rr}(r) + (n-1) r^{n-2} w_r(r) \right) \\ & = 0. \end{aligned}$$

$$\Leftrightarrow \alpha r^{n-1} w(r) - \frac{1}{2} r^n w_r(r) - \left(r^{n-1} w_r(r) \right)_r = 0$$

Take $\alpha = -\frac{1}{2}(n-1)$ Then

$$\alpha r^{n-1} w(r) - \frac{1}{2} r^n w_r(r) = -\frac{1}{2} n \cdot r^{n-1} w(r) - \frac{1}{2} r^n w_r(r)$$

Therefore

$$\left(-\frac{1}{2} r^n w(r) - r^{n-1} w_r(r) \right)_r = 0.$$

$$\Rightarrow \exists a \in \mathbb{R} \text{ s.t. } -\frac{1}{2} r^n w(r) - r^{n-1} w_r(r) = a \quad (*)$$

Assume more that $r^n w(r), r^{n-1} w_r(r) \rightarrow 0$ ($r \rightarrow \infty$).Take $a = 0$ by taking $r \rightarrow \infty$ in $(*)$ From $w_r(r) = -\frac{1}{2} r w(r)$

$$\Rightarrow w_r(r) + \frac{1}{2} r w(r) = 0$$

$$\begin{aligned} \text{multiply } e^{\frac{1}{4} r^2} \Rightarrow & e^{\frac{1}{4} r^2} w_r(r) + \frac{1}{2} r e^{\frac{1}{4} r^2} w(r) = 0 \\ & \quad \quad \quad \left(e^{\frac{1}{4} r^2} \right)_r \end{aligned}$$

$$\Rightarrow \left(e^{\frac{1}{4} r^2} w(r) \right)_r = 0$$

$$\Rightarrow \exists b \in \mathbb{R} \text{ s.t. } e^{\frac{1}{4} r^2} w(r) = b.$$

$$\Rightarrow w(r) = b e^{-\frac{1}{4} r^2}$$

Recall $r = |y|, y = x/\sqrt{t}$.

Then $u(x, t) = t^\alpha v(r)$
 $= t^\alpha w(r)$
 $= t^{-\frac{n}{2}} b \cdot e^{-|x|^2/4t}$
 $\therefore b t^{-\frac{n}{2}} \exp\left(-\frac{|x|^2}{4t}\right)$ is a sol of (12.1)

Def (fundamental sol of the heat eq).

For $x \in \mathbb{R}^n, t \in \mathbb{R}$

$$\Phi(x, t) := \begin{cases} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x|^2}{4t}\right) & t > 0 \\ 0 & t \leq 0 \end{cases}$$

is called a fundamental sol of the heat eq.

<Cauchy problem / Initial value problem>

Consider

$$(12.4) \quad \begin{cases} \frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, & x \in \mathbb{R}^n, t > 0 \\ u(x, t) = u_0(x), & x \in \mathbb{R}^n \end{cases}$$

Here $u_0: \mathbb{R}^n \rightarrow \mathbb{R}^n$ given

$u: \mathbb{R}^n \times [0, \infty) \rightarrow \mathbb{R}$ unknown

$\rightarrow$ We want to represent u by u_0 .

Let define

$$\begin{aligned} u(x, t) &:= \int_{\mathbb{R}^n} \Phi(x-y, t) u_0(y) dy \\ &= \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} \exp\left(-\frac{|x-y|^2}{4t}\right) u_0(y) dy \\ &\quad (\text{for } x \in \mathbb{R}^n, t > 0) \end{aligned}$$

Thm 12.2

Let u_0 be a bounded continuous func. on $\mathbb{R}^n$

Let u be defined by (12.5)

Then we have the following

1) $u \in C^\infty(\mathbb{R}^n \times (0, \infty))$

2) $\frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, \quad \forall x \in \mathbb{R}^n, t > 0$

3) $\forall x_0 \in \mathbb{R}^n, u(x, t) \rightarrow u_0(x_0)$ as $(x, t) \rightarrow (x_0, 0)$
 $t \rightarrow 0^+$

Idea of proof.

1) difficult (Need properties of convolutions).

$$\begin{aligned} 2) \quad \frac{\partial u}{\partial t}(x, t) &= \frac{\partial}{\partial t} \int_{\mathbb{R}^n} \Phi(x-y, t) u_0(y) dy \\ &= \int_{\mathbb{R}^n} \frac{\partial}{\partial t} \Phi(x-y, t) u_0(y) dy \end{aligned}$$

and

$$\begin{aligned} \Delta u(x, t) &= \Delta_x \left(\int_{\mathbb{R}^n} \Phi(x-y, t) u_0(y) dy \right) \\ &= \int_{\mathbb{R}^n} \Delta_x \Phi(x-y, t) u_0(y) dy \end{aligned}$$

Using $\frac{\partial \Phi}{\partial t}(x, t) = \Delta \Phi(x, t)$

then $\frac{\partial u}{\partial t}(x, t) - \Delta u(x, t) = 0, \quad \forall x \in \mathbb{R}^n, t > 0.$

3) We should consider

$$|u(x, t) - u_0(x_0)| \rightarrow 0 \text{ as } (x, t) \rightarrow (x_0, 0)$$

$$|u(x, t) - u_0(x_0)|$$

$$\stackrel{(12.5)}{=} \left| \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y-x|^2}{4t}\right) u_0(y) dy - u_0(x_0) \right|$$

Note that

$$\int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x|^2}{4t}\right) dx = 1.$$

for any $t > 0$ we have

$$\begin{aligned} & |u(x, t) - u_0(x_0)| \\ &= \left| \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y-x|^2}{4t}\right) u_0(y) dy \right. \\ &\quad \left. - \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|y-x|^2}{4t}\right) u_0(x_0) dy \right| \\ &= \left| \int_{\mathbb{R}^n} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) (u_0(y) - u_0(x_0)) dy \right| \\ &\leq \left| \int_{B_\delta(x_0)} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) \underbrace{(u_0(y) - u_0(x_0))}_{\text{small}} dy \right| \\ &\quad + \left| \int_{\mathbb{R}^n \setminus B_\delta(x_0)} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) \underbrace{(u_0(y) - u_0(x_0))}_{\text{small}} dy \right| \end{aligned}$$

$$=: I_\delta + J_\delta \quad (\delta > 0).$$

$$I_\delta : |y - x_0| < \delta \Rightarrow u_0(y) - u_0(x_0) : \text{small} \Rightarrow I_\delta \rightarrow 0.$$

$$J_\delta : |y - x_0| < \delta$$

$$\Rightarrow \exp\left(-\frac{|y-x_0|^2}{4t}\right) : \text{small} \Rightarrow J_\delta \rightarrow 0.$$